

# Franco David Barrionuevo

Master of Spatial Information Applications (MAIE) Student · Córdoba, Argentina

## Contact Details

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**GitHub:** <https://github.com/francobarrionuevoenv21>

## Work Experience

### Scholar

#### Instituto Gulich

02/2026 – Present

Full-time, fully funded scholarship to attend the Master of Spatial Information Applications (MAIE) program courses at the CONAE's Teófilo Tabanera Space Center (CETT).

Research topic: Implementation of an Air Quality Model System, and its Evaluation Using Satellite Data.

### Climate Change Consultant

#### Escuela de Hábitat y Sostenibilidad (EHyS-UNSAM)

07/2025 – 01/2026

Reconstructed the Marine Heatwaves (MHW) historical record from 1980 to 2020 in the Brazilian Current to assess extreme climate events and ecological impacts in the Buenos Aires Coast area.

Implemented large geospatial data processing workflows in Python for a wide variety of ocean data sources: bathymetry, SST interpolated data, ocean reanalysis and atmospheric reanalysis.

### Anaerobic Digestion Technologist

#### Instituto Nacional de Tecnología Industrial (INTI)

06/2023 – 02/2026

Led projects on organic wastes mapping for biodigestion generation using data science and GIS tools.

Developed tailored software in Python to automatize biological laboratory test data processing.

Performed Biochemical Methane Potential (BMP) tests on conventional and non-conventional organic substrates (e.g., microalgae, nopal).

### Research Scholar

#### Universidad Nacional de San Martín (UNSAM)

02/2024 – 08/2025

Implemented geoprocessing tools in Python, Google Earth Engine, and GIS software to analyze and visualize satellite-derived data from various sources.

Reconstructed historical paddy fields expansion implementing a timeseries clustering, evaluated historical human-caused fires, and assessed air quality affection caused by fires through satellite-derived tropospheric air pollutants measurements.

### Impact Scholar

#### Climatematch Academy

10/2023 – 03/2024

Implemented multi-dimensional arrays handling tools in Python for large geospatial data processing.

Programmed algorithms in Python for processing satellite-derived burned area products and climate reanalysis data.

Authored a scientific publication detailing the methodology and results obtained during the project's duration.

## Research Scholar

**Instituto Nacional de Tecnología Industrial (INTI)**

08/2022 – 07/2023

Conducted research on the national availability of organic wastes from industrial processes for utilization in biogas production. Mapped the national distribution of organic waste-producing industries in Argentina. Performed laboratory work, including the characterization of anaerobic sludges and organic substrates, and conducted Biochemical Methane Potential (BMP) tests.

## Education and Training

**Master of Spatial Information Applications (MAIE)**

**Instituto Gulich. Córdoba, Argentina**

02/2026 – In progress

**Courses completed and approved:** Mathematics, Introduction to Remote Sensing, Introduction to intelligent methods for planning, scheduling, and execution problem solving, Statistics, and Numerical Methods Applied to Satellite Data Processing

**Remaining Coursework:** Environmental Remote Sensing, Digital Processing of Satellite Imagery and Geographic Information Systems (GIS), Applications of Synthetic Aperture Radar (SAR) Imagery, Numerical Models for Early Warning Systems, Risk Mapping, and Simulation, Spatial Analysis and Risk Assessment, Environmental Assessment, Monitoring, and Management Tools, Research Methodology and Thesis Development Tools, and Integrative Project.

## Environmental Engineering

**Universidad Nacional de San Martín (UNSAM). Buenos Aires, Argentina**

2017 – 2025

**Courses and topics covered**

**Specialization:** Sanitary and Waste Treatment Engineering.

**Optional courses:** Programming, Data Science and Machine Learning, GIS, and Remote sensing.

## Final Project

**Title:** Prefeasibility Study for the Installation of an Anaerobic Biodigester for the Codigestion of Pig Manure and Nopal (*Opuntia ficus-indica*) in Rosario Vera Peñaloza, La Rioja.

## Medicine

**Universidad de Buenos Aires (UBA). Buenos Aires, Argentina**

2017 – 2025

## Diploma in Data Science and Artificial Intelligence

**Universidad Nacional de San Martín (UNSAM). Buenos Aires, Argentina**

02/2023 – 10/2023

**Artificial Intelligence and Quantum Computing for Earth Observation**

**Gulich Institute, Geoscience and Remote Sensing Society (GRSS) & Pavia University. Córdoba, Argentina**

10/2025

## Introduction to High Performance Computing in Python

**Gulich Institute. Córdoba, Argentina**

09/2025

**Remote Sensing as a Tool for Earth and Forest Resource Monitoring: Uses and Applications of the SAR SAOCOM and Sentinel-1 Missions**

**Gulich Institute, GRSS & European Space Agency (ESA). Córdoba, Argentina**

10/2024

## Language Skills

**Spanish** (Native)

**English** (Reading: C1, Writing: C1, Speaking: B2)

**Portuguese** (Reading: A2, Writing: A2, Speaking: A2)

## Publications

Barrionuevo, F., Rocha, R., Rojas, L., Rengifo, R., & Farfán C. (2024). *Latitudinal Assessment of Burned Area within the Argentinian Andean-Patagonian Forest for the 2001-2019 Fire Seasons*. <https://doi.org/10.5281/zenodo.12206635>

## Conferences and Seminars

### Oral Presentations

#### Geomatics in Environmental Monitoring GEM-24 Workshop

Virtual | 2024

**Title:** *Uso de técnicas de clustering sobre series temporales de índices verdes MODIS para la delimitación de complejos arroceros en el Bajo de los Saladillos (Santa Fe).*

#### Climatematch Impact Scholars Program 2023-2024 Seminars

Virtual | 2024

**Title:** *Latitudinal Assessment of Burned Area within the Argentinian Andean-Patagonian Forest for the 2001-2019 Fire Seasons.*

### Poster Presentations

#### V Simposio de Residuos

Mar del Plata, Argentina | 2025

**Title:** *A Methodology for Mapping and Use in Biogas Production.* **Barrionuevo, Franco;** Videla, Estefanía; Farías, Camila; Flesler, Fanny; Rodríguez, Ruth.

#### Tropentag 2025

Bonn, Germany | 2025

**Title:** *Land use changes and anthropogenic impacts due to rice fields in Santa Fe (Argentina).* Clauser, C. D., Peluso, J., Aronzon, C., **Barrionuevo, F.**, Ali Santoro, V., Morandeira, N.

#### VII Congreso Nacional de Conservación de la Biodiversidad

San Juan, Argentina | 2025

**Title:** *Degradación de las vegas y bosquecitos de chacay por cambios de cobertura e incendios en la cuenca del río Blanco (Mendoza, Argentina).* Torre Azzaroni, María Luisina; **Barrionuevo, Franco David;** Rubio, María Cecilia; Morandeira, Natalia Soledad.

#### Expo EHyS 2024 (EHyS-UNSAM)

Buenos Aires, Argentina | 2024

**Title:** *Enfoque satelital para evaluar quemas en arroceras.* **Barrionuevo, Franco David;** Clauser, Carlos David; Morandeira, Natalia Soledad.

#### IV Simposio de Residuos

Mendoza, Argentina | 2023

**Title:** *Participación en programa internacional de ensayos de aptitud de potencial metanogénico de sustratos.* Videla Estefanía; Camila Farias; Diego Rodriguez; Fanny Flesler; **Franco Barrionuevo;** Ruth Rodriguez.